ZEKA EMO

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Computer Engineering graduate with experience in software engineering, applied AI, and machine learning research at Grab, DGIST, and Diponegoro University. Experienced in preparing datasets, exploring data, engineering features, adapting model architectures, and evaluating predictive performance across computer vision, NLP, graph-based bug localization, and recommendation systems. Research contributions include comparing CNN backbones for plant detection and co-authoring a published SAR ship-detection paper. Combines Python-based experimentation and problem-solving with Go and React development experience to build practical, data-driven solutions.

Work Experiences

Daegu Gyeongbuk Institute of Science and Technology (DGIST) – **Daegu, South Korea** Jun 2026 – Aug 2026
Research Intern
- Conducted research on graph-based bug localization, representing software repositories as graphs to capture structural relationships among code entities and identify potential bug locations.

Diponegoro University – **Semarang, Indonesia** Aug 2025 – Jun 2026
Research Assistant
- Reconstructed the YOLO architecture using C3Ghost modules to optimize lightweight small-object detection in SAR maritime imagery, including dataset preparation and model evaluation.
- Co-authored the full research manuscript for Q3-indexed journal submission, articulating technical insights and results in writing.

Grab – **Jakarta, Indonesia** Sept 2025 – Feb 2026
Full-Stack Engineer Intern
- Built and maintained internal data monitoring tools using Go and React, improving reliability through production debugging, legacy feature migration, and API development.

PT Jaminan Kredit Indonesia - **Bandung, Indonesia** Jul 2024 - Aug 2024
Back-End Developer Intern
- Developed and implemented a comprehensive letter management system, streamlining the tracking of incoming and outgoing correspondence for West Java Regional Office.

LuarSekolah - **Bandung, Indonesia** May 2023 - Jun 2023
Mobile Developer Intern
- Leveraged Flutter and Dart to design and develop a functional Android app prototype, incorporating 5 key widgets and 5 features.

Education Level

Universitas Diponegoro - **Semarang, Indonesia** Aug 2022 - Present
Undergraduate Computer Engineering
- GPA: 3.88/4.00
- Achievements:
  - Best Presentation Team - Bangkit Academy 2024 Batch 2 (National Scale)
  - Third Runner-Up for Paper Competition - AEROSPACE 2024 (National Scale)
  - Second Runner Up for Best Essay Competition - UNLIMITED 2023 (National Scale)

Projects

- **YOLOv5 Backbone Comparison on Plant Detection (2026)**: Reconstructed YOLOv5 with four CNN backbones (EfficientNetB1, ResNet152, VGG16-BN, MobileNet-Large) to identify plant species from field-collected leaf images.
- **SAR Ship Detection Research** (2025): Built and evaluated a lightweight deep learning model by reconstructing YOLO with C3Ghost module for small-target detection in maritime SAR imagery.
- **IoT-Based Essential Oil Distillation System** (2025): Developed the electronic control system and IoT integration for a water-cooling mechanism in an essential oil distillation device, enabling automated temperature regulation and real-time monitoring.
- **Harmful Content Filtering Extension** (2025): Created a Chrome extension that filters toxic comments on Twitter using a fine-tuned IndoBERT model with Hugging Face Inference API integration.
- **AGV Medical Assistant** (2025): Designed a 3D-CNN for human recognition, implemented the A* pathfinding algorithm, and developed a position-tracking database using an Indoor Positioning System (IPS) for an autonomous guided vehicle (AGV) robotics system.This project aimed to reduce the workload of health workers.
- **SiBantu** (2024): A location-based service platform utilizing machine learning for matching users with home repair service providers.
- **Comparison Machine Learning Methods in Plant Recommendation System** (2024): Demonstrated that supervised learning approaches are most suitable for enhancing recommendation accuracy. Presented findings and methodology at AEROSPACE 2023.
- **DF Seperator** (2023): A conceptual waste separator system featuring Waste Image Recognition powered by a Data Fusion algorithm. This project showcased proficiency in computer vision and aimed to improve waste sorting efficiency.
- **Plant Recommendation** (2023): An integrated multi-sensor data and machine learning to provide tailored plant cultivation advice.
- **Recreation Recommendation Program** (2023): Develop a Recreation Recommendation system to help tourists exploring local landmarks using Python.

Organizational Experience

TEDxUNDIP 4.0 Oct 2024 - May 2025
Project Manager
- Directed TED Talks main event, orchestrating all speaker sessions, entertainment segments (traditional dance, theater), and event rundown.
- Formulated the concept and managed the exhibition featuring 5 sensory-cultural themed booths.

Certificates

- **Advanced Learning Algorithm (2024)**: Deeper exploration of optimization and ML algorithms.
- **Unsupervised Learning: Recommenders, Reinforcement Learning (2024)**: Covers clustering, recommender systems, and basics of Reinforcement Learning.
- **Convolutional Neural Network in TensorFlow (2024)**: Building and training CNNs for computer vision.
- **Natural Language Processing in TensorFlow (2024)**: NLP pipelines using TensorFlow.
- **Sequence, Time Series, and Prediction (2024)**: Time series modeling and forecasting techniques.
- **Generative AI for Everyone (2024)**: Overview of generative AI applications and implications.
- **Custom and Distributed Training with TensorFlow (2024)**: Scaling ML models with TensorFlow strategies.
- **Advanced Computer Vision with TensorFlow (2024)**: High-level vision tasks and architectures.
- **Build Basic Generative Adversarial Networks (GANs) (2024)**: Introductory GANs for image generation.
- **Custom Models, Layers, and Loss Functions with TensorFlow (2024)**: This course covers custom models, layers, and loss functions using TensorFlow.
- **Data Pipelines with TensorFlow Data Services (2024)**: This course focuses on building data pipelines with TensorFlow Data Services.

Language Proficiency Test

- **TOEFL** (2025): Listening Comprehension: 63, Structure and Written Expression: 52, Reading Comprehension: 67, Total Scores: 607
- **IELTS** (2022): Listening: 7.5, Reading: 8.0, Writing: 6.0, Speaking: 6.0, Overall Band Score: 7.